

SURYANSHU CHOUDHARY

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Education

Indian Institute of Information Technology, Ranchi

2023 – 2027

B.Tech., Electronics and Communication Engineering (IoT and Embedded Systems)

CGPA: 8.00

Experience

UpstageX Private Limited

May 2026 – Present

Co-founder & Engineering Lead

Remote

- Co-founded an AI-powered influencer marketing and UGC platform and shipped it to production: creator discovery and matching across niches, automated contract generation with milestone-based INR escrow, and AI-generated campaign briefs with deliverable tracking and approval workflows.
- Largest contributor to the TypeScript codebase across a six-person engineering team, working end to end across the discovery, contracts and escrow, and campaign workflow surfaces.
- Incorporated the company and ran cold outbound to consumer brands, drawing replies and inbound interest from Mamaearth (Honasa Consumer), CASHe, and Ideal Kitchen.

Projects

Suhani | Kotlin, Android, Sarvam AI, ONNX, SQLite, Python

2026

- Built a real-time Hindi/Hinglish voice assistant for a home tailoring business: one button, spoken orders and due dates go into a ledger, answerable by voice. Shipped as an Android app in daily production use by a non-technical user, running entirely on-device apart from three model API calls.
- Engineered the full duplex voice loop in Kotlin: on-device Silero VAD and a speaker gate so side conversation in the room never triggers a turn, Sarvam Saaras v4 streaming STT (Hinglish code-mix), a Sarvam-105B tool-calling brain, and Bulbul v3 streaming TTS over a local SQLite ledger with transliteration-aware fuzzy client matching.
- Chose the model configuration empirically rather than by taste: built a bake-off harness and scored roughly 1,200 calls, landing on a setup that cut time-to-first-token to 1.0–1.3s, and moved date arithmetic out of the model into a deterministic tool after scoring exposed invented due dates.
- Hardened it for an operator who cannot debug it: every ledger write carries an idempotency guard because the model multi-fires tool calls, prompts are regression-tested at 300/304 against a Python harness, and 56 pure-JVM unit tests gate every build through a SQL driver seam that needs no emulator.

BitTorrent Client | Node.js, Electron.js | GitHub

April 2025

- Implemented the BitTorrent protocol from scratch in JavaScript with zero native modules: bencode encoding/decoding, tracker communication, peer handshake, and the peer wire message protocol.
- Downloaded pieces concurrently from multiple peers with per-piece SHA-1 integrity verification, and seeded completed files back into the swarm.
- Packaged as a cross-platform desktop client in Electron.js with a live view of connected peers and transfer state.

MeshSync | C#, QML, Bluetooth LE, Wi-Fi | GitHub

August 2026

- Built a local-first universal clipboard across devices over Bluetooth LE and Wi-Fi with no cloud, server, or account.
- Implemented peer-to-peer syncing with no host/client asymmetry, securing connections with keypair device authentication and generating a fresh AES-256 key per connection.

Achievements

- 1st Runner-up**, HyperVerge Hackathon 2024. Built Widget Manager as a two-person team: a Chrome new-tab extension with 14+ productivity widgets including a Pomodoro timer and Spotify player (React, Google Apps Script).
- 2nd Runner-up**, Quasar 2.0 Hackathon, IIIT Ranchi 2025, from 48 qualified teams, for a Puppeteer crawler that walks an entire site and reports every dead link (GitHub).

Technical Skills

Languages: Python, TypeScript, JavaScript, Kotlin, C, C++, C#, SQL, Embedded C

Frameworks: React, React Native, Android, Node.js, Express.js, FastAPI, Tailwind CSS

Data & Infrastructure: PostgreSQL, MongoDB, SQLite, Redis, Firebase, Docker, Git, AWS

AI & Voice: LLM tool-calling, streaming STT/TTS, on-device VAD (ONNX), prompt evaluation harnesses

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Electronics & Embedded: Arduino, Embedded C, Sensor Interfacing (MQ, DHT11, IR), UART / I2C / SPI